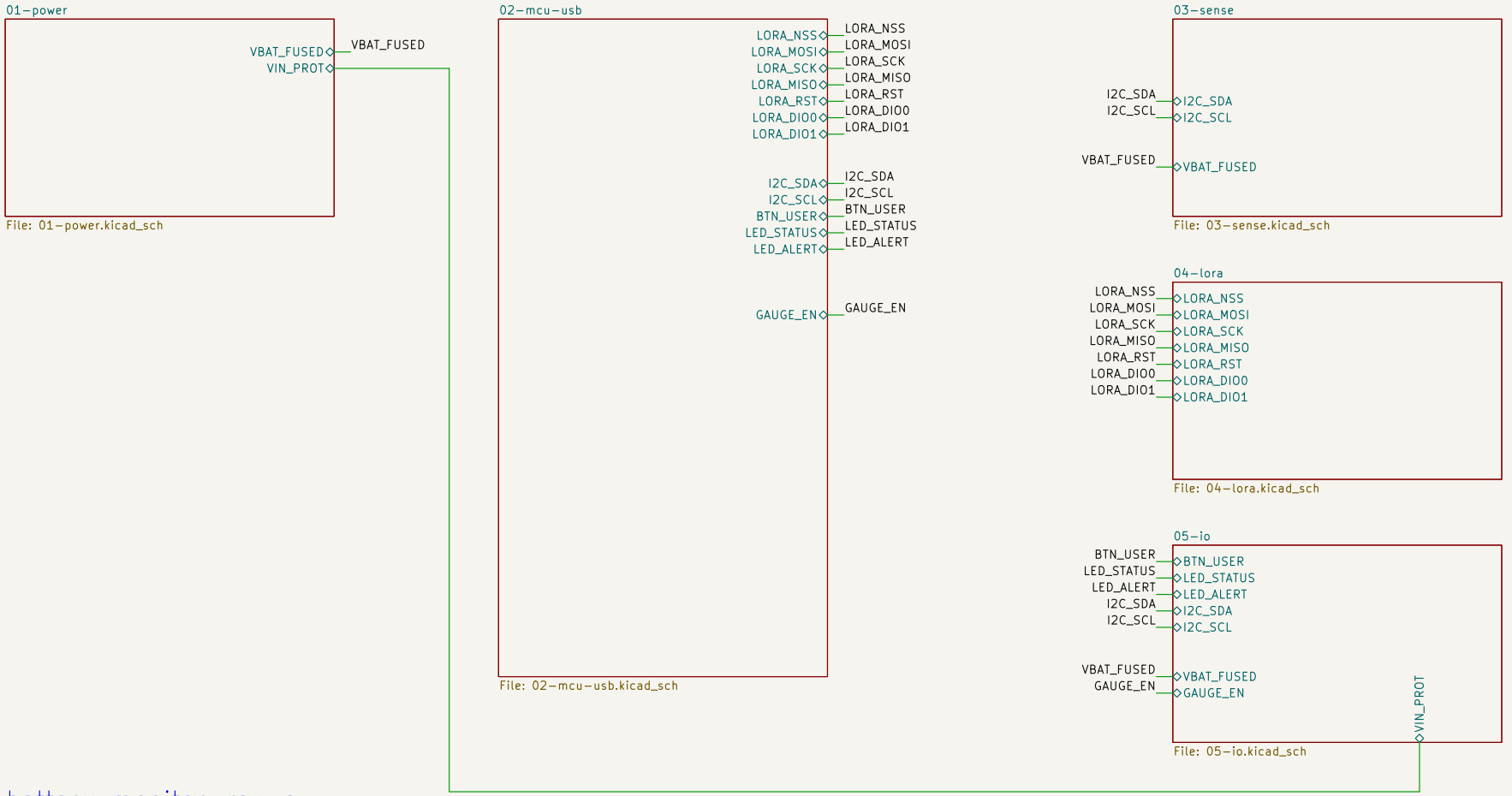


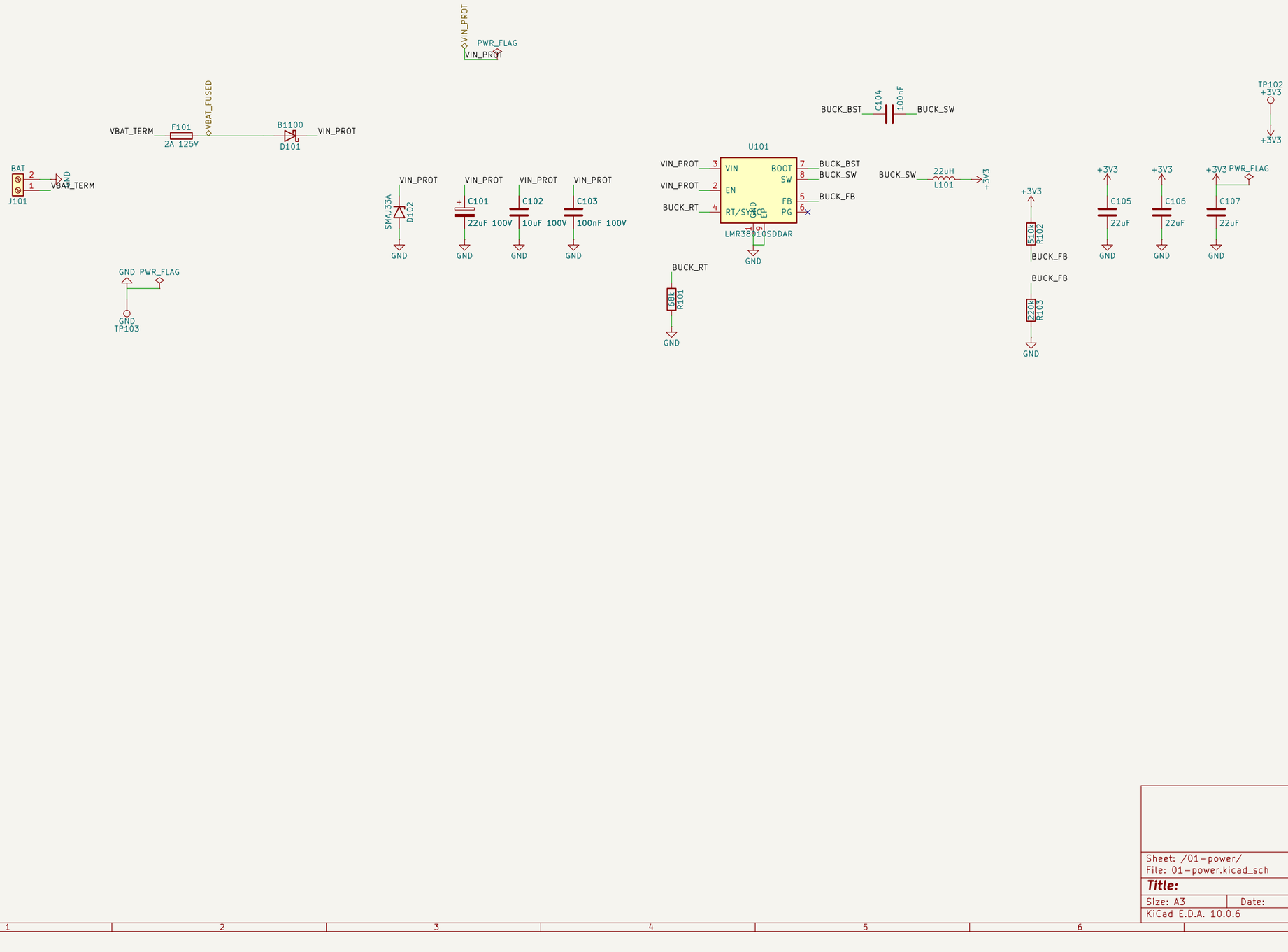
battery−monitor−rev−a

Revision: rev−a Date: 2026−10−18 Status: Reworked 4−layer design (62 parts)
ESP32−S3−WROOM−1−N16R8 battery voltage monitor, systems up to 24 V,
Wi−Fi reports, Ra−02 433 MHz LoRa fitted (as rev−c).
01−power: inlet protection, LMR38010SDDAR 3.3 V buck
02−mcu−usb: native USB−C, buttons/UART removed 04−lora: inherited from rev−c
03−sense: always−on 0.1 % divider, ADS1115 (I2C 0x48)
05−io: external button JST−XH 2P (J504), status/alert LEDs, I2C pull−ups, LED voltmeter JST−XH 3P (J503)
Design record: evidence/rework−20261018/step1−sch/CHANGES.md

Sheet: /		
File: battery−monitor−rev−a.kicad_sch		
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01—POWER: battery inlet protection and 3.3 V buck straight from the monitored battery (6—48 V systems, see report section 3).

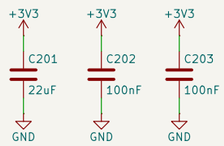
Chain: J101 BAT+ (solder pad 1; two wires from a chassis—mounted XT30, + = pin 1) —> F101 2 A 63 V fast fuse —> VBAT_FUSED —> D101 SS110 series Schottky (reverse polarity, 100 V) —> VIN_PROT:
D102 SMAJ33A TVS + C101 22 uF/100 V electrolytic (damping) + C102 10 uF/100 V + C103 100 nF/100 V —> U101.
Owner decision Q1: systems up to 24 V nominal (31 V equalisation). Documented alternate for 48 V only (not built): SMAJ58A/SMAJ64A, a 125 V fuse, 100 V switch MOSFETs.
Q10: battery leads with alligator clips plug into the chassis XT30 (soldered to J101 pads, 2026—10—15); hot—plug and reversed clips are expected: D101 blocks reversal, D102/C101 handle hot—plug ringing.
Sense tap: VBAT_FUSED (after F101, ahead of D101; Kelvin track from the F101 pad) feeds the always—on divider on 03—sense; the D101 drop never enters the ADS1115 reading.
VIN_PROT (after D101, reverse—safe) feeds the LED voltmeter switch on 05—io (W23, 2026—10—15).
U101 LMR38010SDDAR (PFM + spread spectrum; FDDAR/FSDDAR are forced—PWM): VIN 4.2—80 V (85 V abs), 1 A; EN tied to VIN; RT 68k (E24) = 388 kHz, full regulation to 65 V;
L101 22 uH; FB 510k/220k (E24) —> 3.318 V, no CFF; PG not used. No 5 V rail. Test points: TP102 +3V3, TP103 GND.

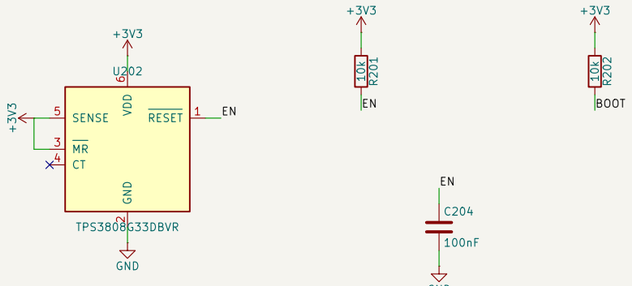


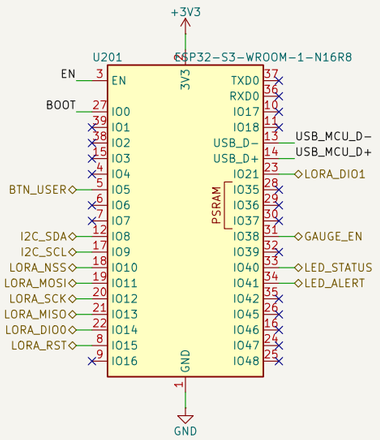
Sheet: /01—power/		
File: 01—power.kicad_sch		
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Size: A3	Date:	**Rev:**
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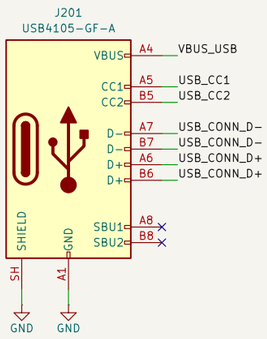
02-MCU-USB: ESP32-S3-WROOM-1-N16R8, EN/RESET, BOOT, USB-C data, UART0. Inherited from antenna-controller-rev-c 02-mcu-usb (owner-approved).

EN: R201 10k pull-up + C204 100nF, U202 TPS3808G33 (3.07 V, open drain, 20 ms with CT open) drives CHIP_PU. SW201 RESET removed per rework 2026-10-18.
BOOT: GPIO0 R202 10k pull-up to +3V3. SW202 BOOT removed per rework 2026-10-18. Straps GPIO3/45/46 isolated (no-connect): GPIO35/36/37 octal PSRAM, no-connect.
USB-C J201 data only: CC1/CC2 separate 5.1k, U203 ESD, 22R series to GPIO19/20; VBUS feeds U203 pin 5 and C205 only (board powered from the battery).
Programming and serial debug use native USB (J201 USB-C USB_Serial_JTAG with built-in auto-reset/auto-download).
UART0 header J202, R207, R211 removed per rework 2026-10-18; TXD0 (IO43), RXD0 (IO44), IO2, IO6 are no-connect.

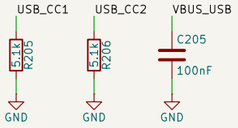










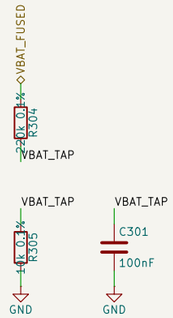


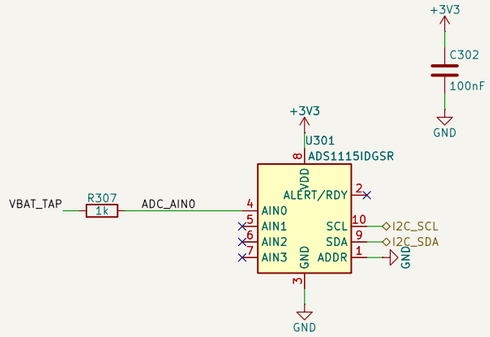
Pin map (battery-monitor-rev-a, reworked 2026-10-18): IO8 I2C_SDA, IO9 I2C_SCL (on-board ADS1115 0x48), IO5 BTN_USER (J504 external button), IO38 GAUGE_EN (LED voltmeter switch on 05-io), IO40 LED_STATUS, IO41 LED_ALERT, LoRa IO10-15/21 as rev-c. Programming/debug: native USB (USB_Serial_JTAG on J201; SW201, SW202, J202 removed).
IO4 no-connect; divider always on, about 0.1 mA (was SENSE_EN). IO1 no-connect. IO7 no-connect (ADS1115 ALERT/RDY not used, firmware polls).
IO2, IO6, TXD0 (IO43), RXD0 (IO44) no-connect (J202 and J302 removed). Free (no-connect): IO1/2/4/6/7/15/17/18/39/42/43/44/47/48.
U201 BOM: ESP32-S3-WROOM-1-N16R8 from owner stock, or N16/N8 (same footprint; PSRAM not used by the firmware).

Sheet: /02-mcu-usb/		
File: 02-mcu-usb.kicad_sch		
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03-SENSE: battery voltage front end: always-on 0.1 % divider on VBAT_FUSED, ADS1115 16-bit ADC (owner Q2).

Divider (W23, 2026-10-15): R304 220k / R305 10k, 0.1 % 25 ppm thin film, straight on VBAT_FUSED (after F101, ahead of D101), always on: about 0.1 mA at 24 V, 0.05 mA at 12 V (the battery feeds a CCTV camera continuously, owner Q8: no divider switch). Ratio 23.0: 0-69 V -> 0-3.0 V: one-point calibration in firmware (NVS).
U301 ADS1115 at I2C 0x48: AIN0 = VBAT_TAP via R307 1k (PGA auto-range: 4.096 V FS -> 2.9 mV, 2.048 V -> 1.4 mV per LSB at the battery).
ALERT/RDY not connected: firmware polls the conversion (single-shot, OS bit of the config register).
AIN1, AIN2, and AIN3 are unused (no-connect, allowed by SBAS444E 9.1.4; J502, R309, R310 removed per rework 2026-10-18).
Reversed alligator clips: the divider tap goes to about -1.4 V at -32 V; the 9.6k divider source plus R307 1k keep the ADS1115 AIN0 clamp current below 0.15 mA (10 mA limit).
No NTC and no current sense (owner Q4, Q6): future options only.

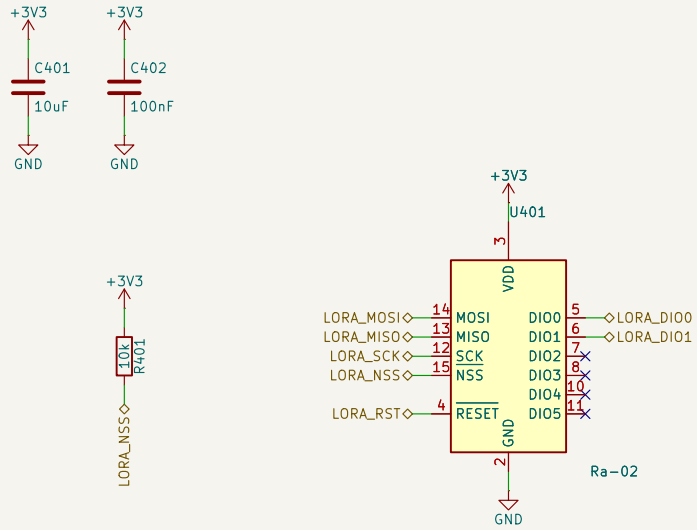




Sheet: /03-sense/		
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04-LORA: Ra-02 433 MHz, inherited from antenna-controller-rev-c sheet 03-lora (pin map identical: FSPI GPIO10 NSS, 11 MOSI, 12 SCK, 13 MISO; DIO0 GPIO14, DIO1 GPIO21, RST GPIO15).

FITTED (owner Q3): Ra-02 on the board exactly as rev-c so the project can grow into LoRa; Wi-Fi stays the report path. Firmware puts the SX1278 to sleep while LoRa is unused.

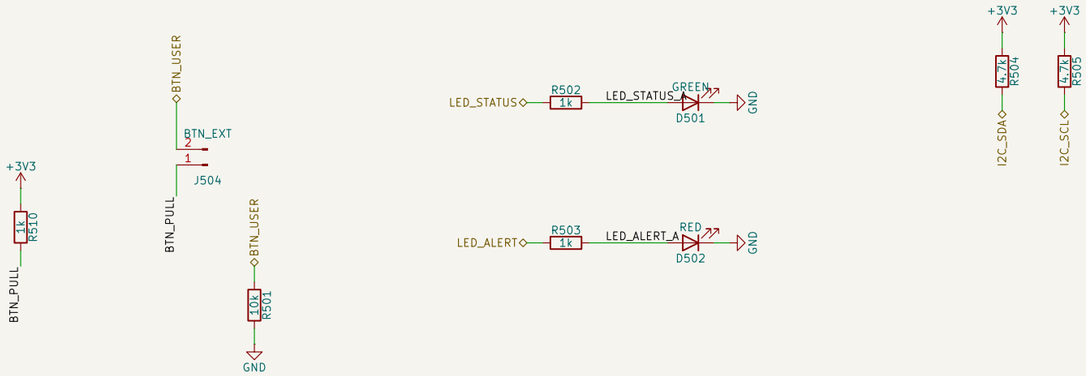


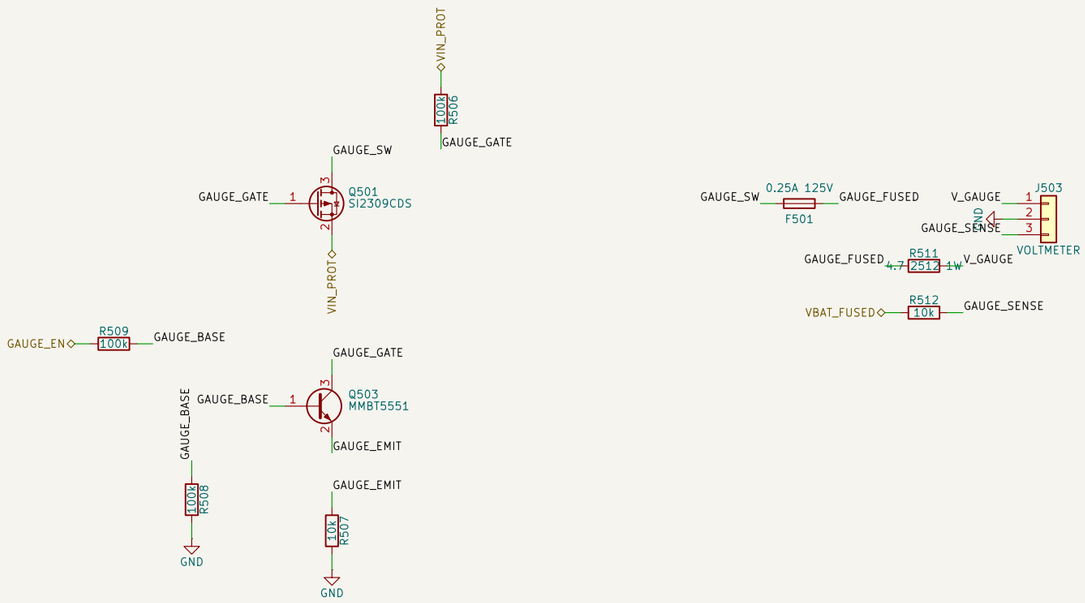
NRESET: no board resistor (SX1278 NRESET floats during POR); firmware: GPIO15 input (high-Z); reset = drive low >= 100 us, then input again, wait 5 ms (Semtech SX1276/77/78/79 7.2.2).
NSS 10k pull-up (R401). SPI start 4 MHz, max 10 MHz. Never transmit without antenna or 50 ohm load; +20 dBm only after VSWR <= 3:1 is proven.
Firmware must put the SX1278 into Sleep (0.2 uA) when unused: after POR it idles in Standby (about 1.6 mA), which would dominate the board standby budget.

Sheet: /04-lora/		
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05-IO: user button, status and alert LEDs, I2C header, spare header, LED voltmeter output.

External wake button J504 (B2B-XH-A 2P) to +3V3 via R510 1k with R501 10k pull-down to GND keeps the DevKit firmware logic (digitalRead HIGH = pressed);
1 BTN_PULL (+3V3 via R510 1k), 2 BTN_USER (GPIO5). GPIO5 is an RTC GPIO: ext1 wake-up from deep sleep works (SW501 replaced by J504 per rework 2026-10-18).
LEDs: D501 (FYLS-0603GD, ~1.2 mA from GPIO40 via R502 1k), D502 (KT-0603R, ~1.4 mA from GPIO41 via R503 1k). No power-up glitch.
I2C bus GPIO8 SDA / GPIO9 SCL: on-board 4.7k pull-ups R504/R505 kept for U301 ADS1115 (0x48); J501 and J502 removed per rework 2026-10-18.





Voltmeter output (owner Q5; simplified W23 2026-10-15; sense resistor 2026-10-15): 0.28 in LED voltmeter 0-100 V. Single P-MOS high-side switch Q501 SI2309CDS, source on VIN_PROT (after D101: reversed clips never reach the module). GAUGE_EN (GPIO38) high -> R509/R508 divider -> Q503 sinks 70-105 uA (R507 10k) -> R506 100k sets VGS 7-10.5 V (limited to VIN_PROT; >= 4 V at a 5.25 V battery). Off: R508 holds Q503 off at reset. J503: 1 V_GAUGE (module supply via F501 0466.250 + R511 4.7 ohm anti-surge), 2 GND, 3 GAUGE_SENSE. SENSE through 10 k from VBAT_FUSED; 2-wire modules use pins 1-2. R512 10k: 3-wire modules read the battery after F101 (no D101 drop); a short on the sense lead is limited to 3.1 mA (96 mW at 31 V). The ADS1115 reading is the reference. Fault coordination (review R2-01): a short on pin 1 is limited by R511 + F501 + Q501; F501 opens in <= 0.2 s (>= 300 %) or 0.04-0.3 ms (adiabatic). R511 fault energy at 5.25 V is <= 12.2 J (2.45 W for <= 5 s), below its 13.2 J overload test point (fixing doc-audit items D2, D25). FIRMWARE: never assert GAUGE_EN above 30.0 V (read the ADS1115 first; module limit 30 V). J504 external button is fed through R510 1k: pressed level 3.0 V, and the GPIO5 power-up low glitch cannot short +3V3.

Sheet: /05-io/
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